

Instructions to the Students:

1. All Questions are Compulsory
2. Draw neat diagram wherever necessary.
3. Figures to right indicates full marks
4. Assume suitable data wherever necessary and mention it clearly

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | |
|---|----|
| A) What is Computer Vision and state Important application of Computer vision | 06 |
| B) Explain CWT for stationary and non-stationary signal | 06 |
| C) What are the Basic parts of Image Formation Process | 06 |

Q.2 Solve Any Two of the following.

- | | |
|--|----|
| A) Explain the terms SNR and PNSR for Image Quality Measurement | 06 |
| B) What are the three types of segmentation? Explain them | 06 |
| C) Applying the region splitting and merging on the following Image will lead to how many isolated pixels for a threshold value ≤ 1 | 06 |

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 \\ 3 & 1 & 6 & 6 \\ 1 & 1 & 5 & 6 \end{bmatrix}$$

Q. 3 Solve Any Two of the following.

- | | |
|---|----|
| A) Explain the color feature and explain the concept edge detection | 06 |
| B) Explain the term Image Scaling https://www.batuonline.com | 06 |
| C) Explain the frequency domain technique for Image Enhancement. | 06 |

Q.4 Solve Any Two of the following.

- | | |
|---|----|
| A) Describe how grey level histograms can be used for Image Segmentation. | 06 |
| B) What are the steps of Image Analysis? Explain with Diagram | 06 |
| C) A 3-bit image patch is given below- | 06 |

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \\ 7 & 2 & 1 \end{bmatrix}$$

Using bit plane slicing the patch is disintegrate into bit planes and the LSB plane is discarded find the reconstructed image patch.

Q. 5 Solve Any Two of the following.

- | | |
|---|----|
| A) Explain the concept of Texture Analysis in details. | 06 |
| B) Explain the concept of Video Surveillance System. | 06 |
| C) What are the Components of Pattern Recognition System? | 06 |

*** End ***